Machine Intelligence 1

2.1 Elements of Statistical Learning Theory

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Background

Central question

What must one know a priori about an unknown functional dependency in order to estimate it on the basis of observations? (Vapnik, 1995)

- Regularization principles for solving ill-posed problems (e.g. Tikhonov)
- Nonparametric statistics (e.g. Parzen)
- Law of large numbers in function space (Vapnik & Chervonenkis)
- Algorithmic complexity (Kolmogorov, Solomonoff, Chaitin)

Empirical Risk Minimization (ERM)

$$E^G_{[\underline{\mathbf{w}}]} = \int P_{(y_T,\underline{\mathbf{x}})}\, e_{(y_T, y(\underline{\mathbf{x}};\underline{\mathbf{w}}))}\, d\underline{\mathbf{x}}\, dy_T \stackrel{!}{=} \min_{\underline{\mathbf{w}}} \quad \text{(generalization error)}$$

mathematical expectation

$\Downarrow$ ERM $\Downarrow$ $\leftarrow$ When does this work?

empirical average

$$E^T_{[\underline{\mathbf{w}}]} = \frac{1}{p} \sum_{\alpha=1}^{p} e_{(y_T^{(\alpha)},\, y(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}}))} \stackrel{!}{=} \min_{\underline{\mathbf{w}}} \qquad \text{(training error)}$$

The law of large numbers

"The sequence of means converges to the expectation of a random variable (if it exists) as the number p of samples increases."

- Can only be applied if set of predictors has a finite number of elements (use Hoeffding's inequality for bound on deviations).
- Extension to infinite sets (e.g. MLPs) is necessary.

Statistical learning theory

Goal: Estimate a desired function from

- a wide set of functions
- a limited number of examples

Achievements of statistical learning theory

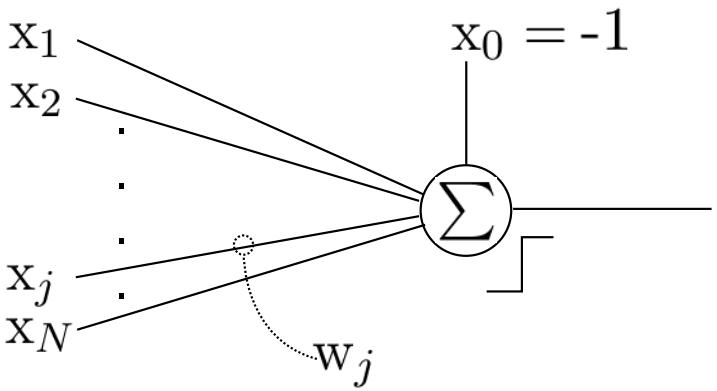
1. formulation of conditions under which ERM works
2. bounds describing **generalization ability** of ERM
3. inductive inference for **small sample sizes** based on these bounds
4. methods for implementing this new type of inference ($\rightarrow$ **SVMs**)

2.1.1 Linear Classifiers: An Example

Linear classifiers

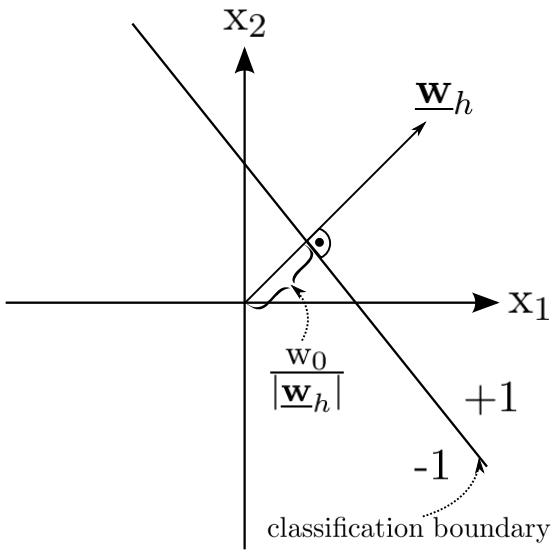
- **data representation:** samples $\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N$ and classes $y_T^{(\alpha)} \in \{-1, 1\}$
- **model class:** binary connectionist neuron

binary connectionist neuron



$$y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) = \mathrm{sign}(\underline{\mathbf{w}}^\top \underline{\mathbf{x}})$$

classification boundary



$$\underline{\mathbf{w}}^\top \underline{\mathbf{x}} = 0$$

Optimization with ERM

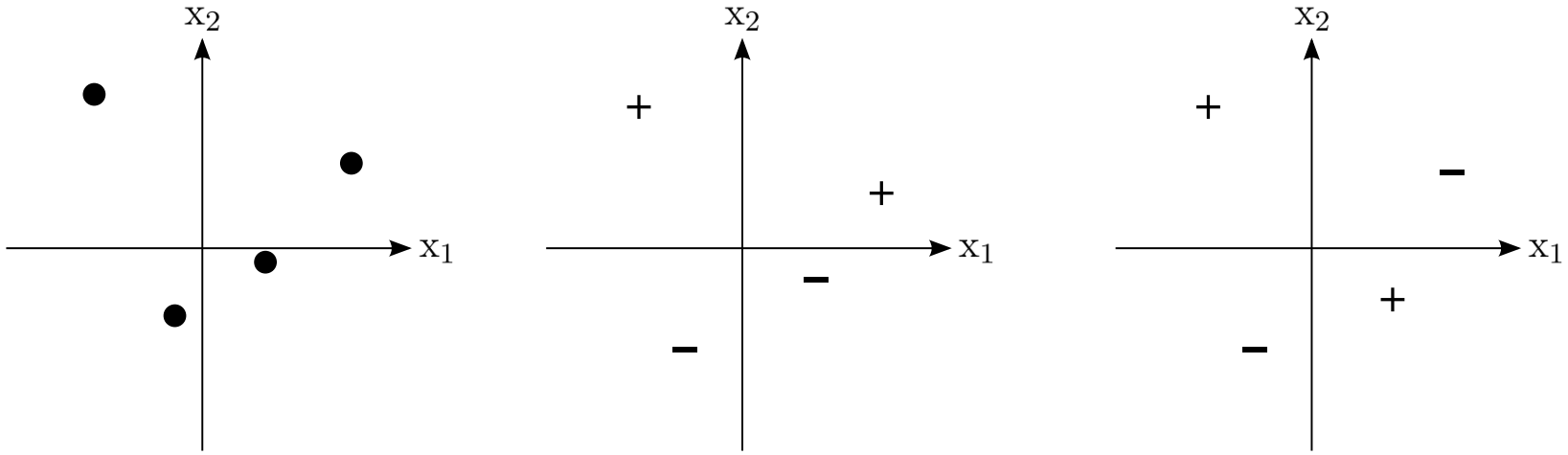
- **performance measure:** classification error
- **optimization:** minimization of the training error (ERM)
  - For which problem setting can we expect small training errors?
- **validation:** evaluation of the prediction error on unseen data
  - For which problem setting can we expect small generalization errors?

Linear separability

$\rightsquigarrow$ classes do not overlap
$\rightsquigarrow$ classes can be separated by a hyperplane

} all data points can be correctly classified by at least one classifier from the set

- observations $\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N, \alpha = 1, \ldots, p$, in *general location* ($\rightarrow$ each subset of N points should be linearly independent)



in general location | separable assignment | assignment not linearly separable

- 14 linearly separable configurations, 2 configurations not separable

Statistics of linear separability (Cover, 1965)

1. **number** $C_{(p,N)}$ of linearly separable assignments $y_T : \underline{\mathbf{x}} \rightarrow y_T$:

$$C_{(p,N)} = 2 \sum_{k=0}^{N-1} \binom{p-1}{k}$$

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2. **fraction** $\Pi_{(p,N)}$ of linearly separable assignments (total number of assignments: 2^p):

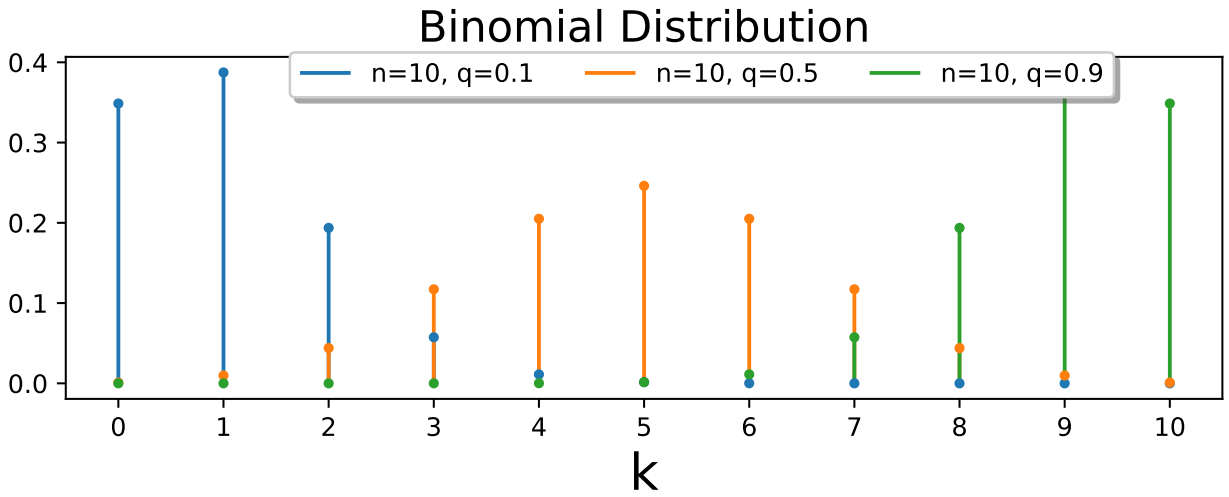
$$\Pi_{(p,N)} = \frac{C_{(p,N)}}{2^p} = \frac{1}{2^{p-1}} \sum_{k=0}^{N-1} \binom{p-1}{k}$$

Asymptotic distribution

$$\Pi_{(p,N)} = \sum_{k=0}^{N-1} \overbrace{\binom{p-1}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{p-1-k}}^{f(k\,|\,p-1,\frac{1}{2})}$$

$$p-1 \to n, \quad \frac{1}{2} \to q$$

$$f(k|n,q) = \binom{n}{k} q^k (1-q)^{n-k} \qquad \text{(Binomial distribution)}$$



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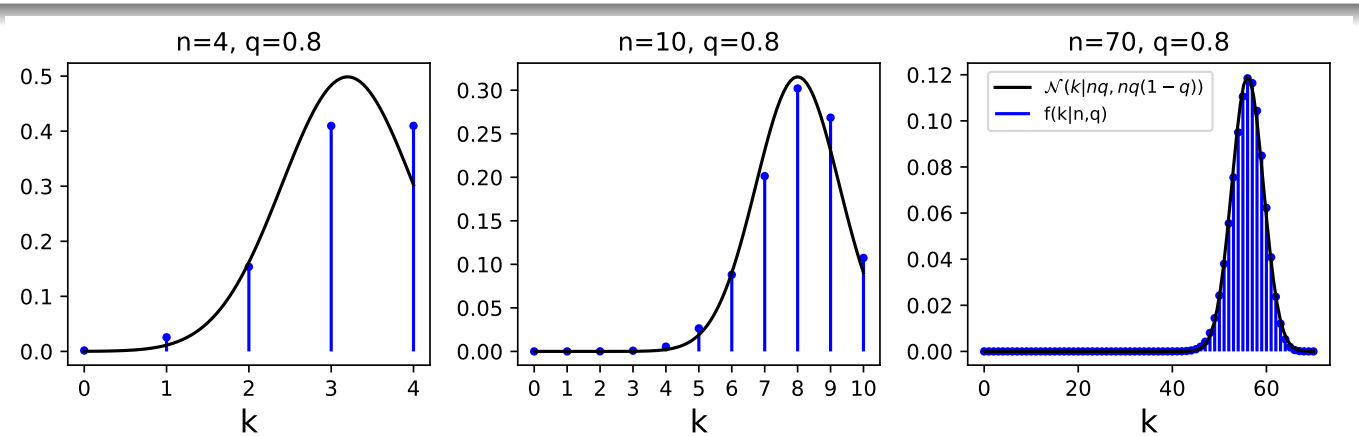
For $n \to \infty$ the Binomial distribution converges to a Gaussian distribution

$$f(k\,|\,n,q) \underset{n\to\infty}{\longrightarrow} \mathcal{N}(k\,|\,nq,\, nq(1-q)) \qquad \text{(asymptotic distribution)}$$



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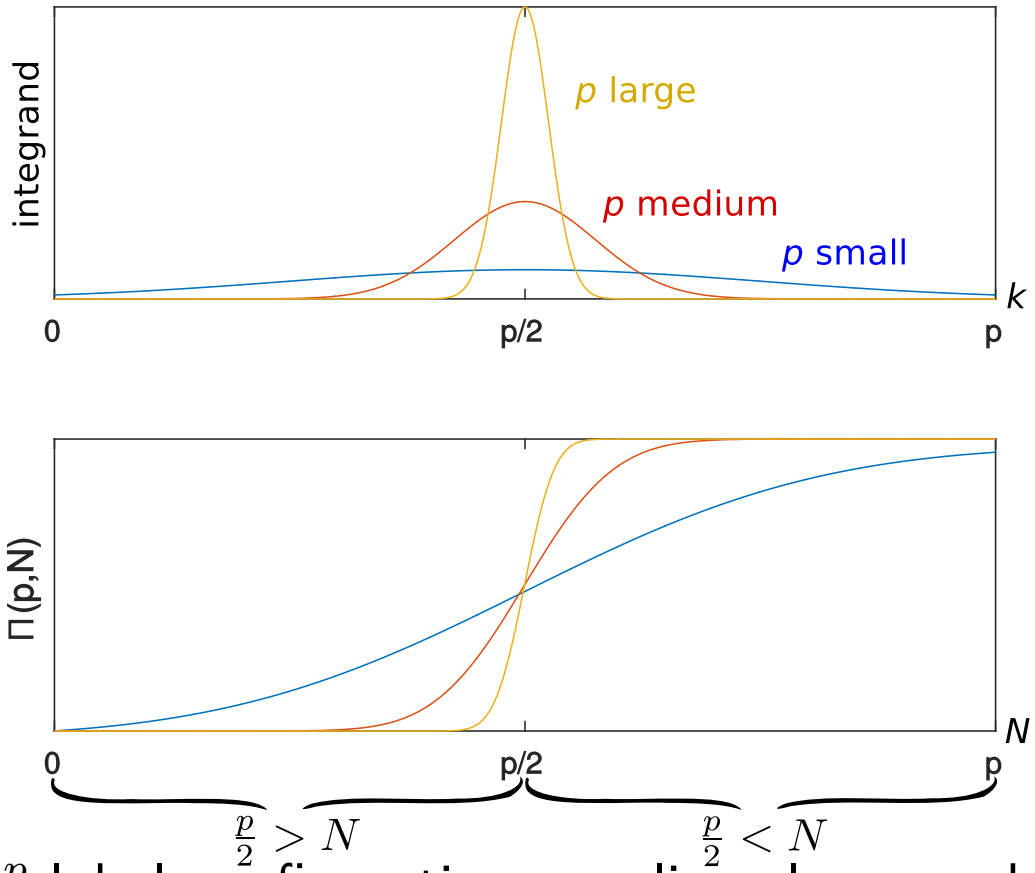
$\Pi_{(p,N)}$ is the cumulative density function of the Binomial distribution $f(k\,|\,p-1,\frac{1}{2})$

In the limit $N, p \to \infty$, with $\frac{p}{N} = \text{const}$, one obtains:

$$\Pi_{(p,N)} \approx \sum_{k=0}^{N-1} \mathcal{N}(k\,|\,\tfrac{p-1}{2}, \tfrac{p-1}{4}) \propto \int_{-\infty}^{N} dk\, \exp\Big(-\tfrac{1}{p/2}(k-\tfrac{p}{2})^2\Big)$$

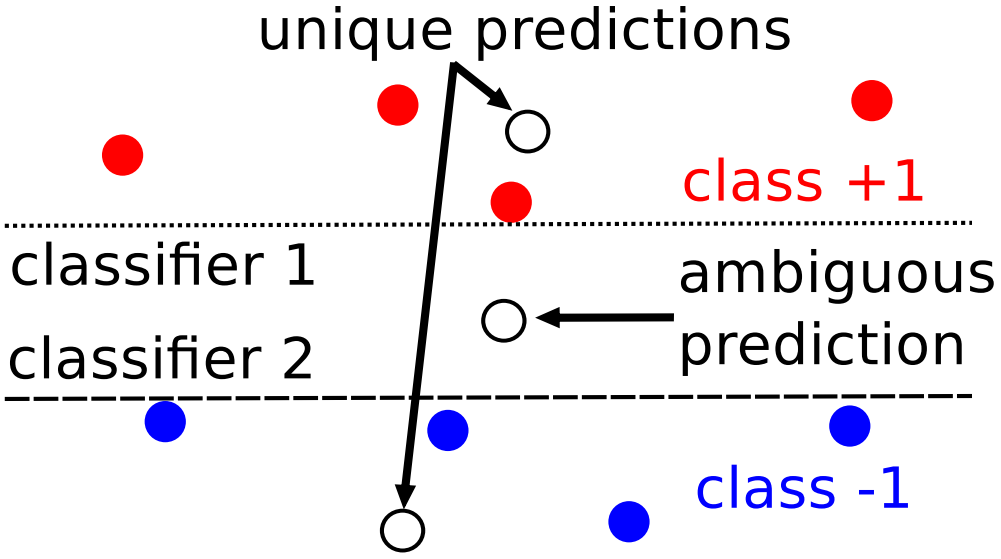
Capacity of the set of linear classifiers

$$\Pi_{(p,N)} \quad \propto \quad \int_{-\infty}^{N} dk \, \exp\left(-\frac{1}{p/2}(k-\frac{p}{2})^2\right)$$



- Almost all 2^p label configurations are linearly separable below $\beta := \frac{p}{N} = 2$.
- β can be understood as a "natural" capacity measure.

Generalization performance (1)



Number of linearly separable assignments on $p-1$ data points, where there are still two choices left for the remaining data point (Cover, 1965):

$$C_{(p,N-1)}$$

Generalization performance (2)

- number of ambiguous, linearly separable assignments: $C_{(p,N-1)}$
- fraction of ambiguous, linearly separable assignments

$$g_{(p,N)} := \frac{\text{ambiguous assignments}}{\text{all separable assignments}} = \frac{C_{(p,N-1)}}{C_{(p,N)}}$$

- if fraction of ambiguous assignments is large, predictions for new data points will not be informative

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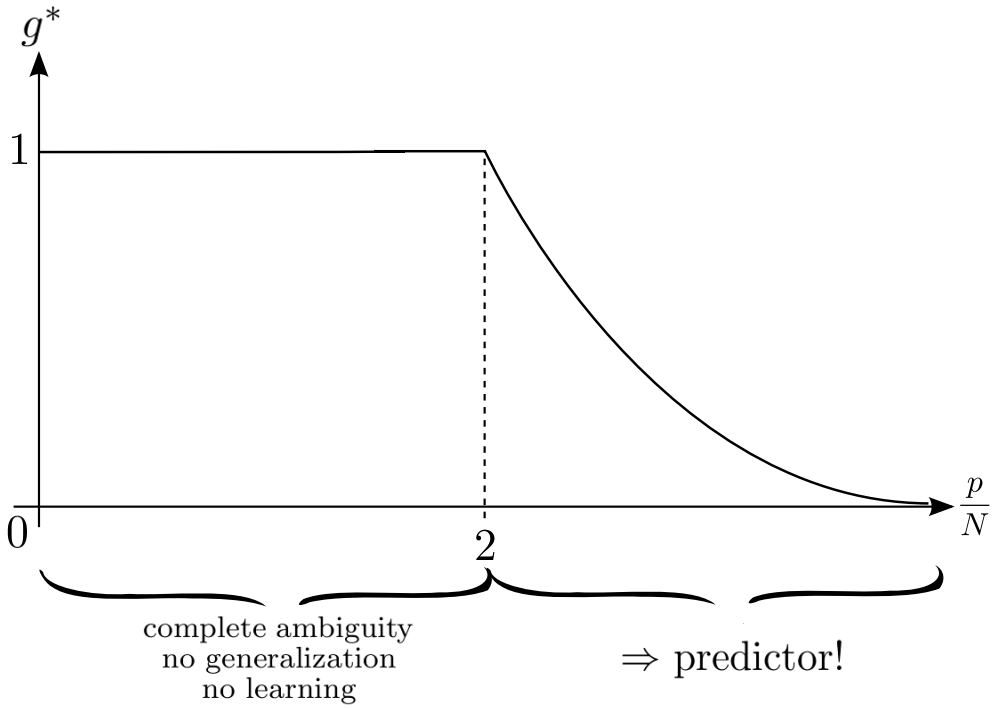
- if fraction of ambiguous assignments is large, predictions for new data points will not be informative
- asymptotic result (Cover, 1965):

$$g^*_{(\frac{p}{N})} = \lim_{\substack{p,N\to\infty,\\ \frac{p}{N}=\text{const}}} g_{(p,N)} = \begin{cases} 1, & \text{for } 0 \leq \frac{p}{N} \leq 2 \\ \frac{1}{\frac{p}{N}-1}, & \text{for } \frac{p}{N} > 2 \end{cases}$$

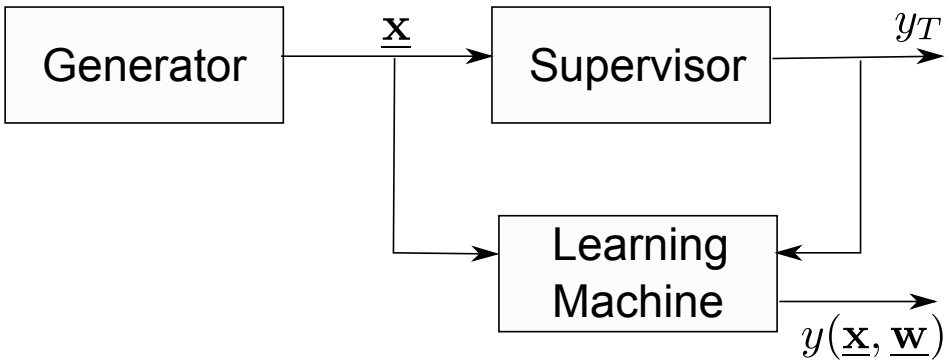
Generalization performance (3)



- For $p/N \leq 2$ almost all predictions are ambiguous.
- For $p/N > 2$ the classifier is able to generalize to new data points.

2.1.2 Formulation of the Inductive Learning Problem

Inductive learning for binary classification



- generator draws $\underline{\mathbf{x}} \in R^N$ from unknown $P(\underline{\mathbf{x}})$
- supervisor returns output $y_T \in \{-1, +1\}$ to every $\underline{\mathbf{x}}$ from unknown $P(y_T|\underline{\mathbf{x}})$
- learning machine estimates function $y(\underline{\mathbf{x}}, \underline{\mathbf{w}}) \in \{-1, +1\}$ by selecting $\underline{\mathbf{w}} \in \Lambda$ from the set Λ of possible parameter combinations
- performance is measured with 0-1 loss function

$$e_{(y_T, \underline{\mathbf{x}}; \underline{\mathbf{w}})} \quad = \quad \frac{1}{2}\left(1 - y_T \cdot y(\underline{\mathbf{x}}, \underline{\mathbf{w}})\right)$$

Empirical Risk Minimization (ERM)

$$e_{(y_T,\underline{\mathbf{x}};\underline{\mathbf{w}})} \quad = \quad \tfrac{1}{2}\big(1 - y_T \cdot y(\underline{\mathbf{x}},\underline{\mathbf{w}})\big)$$

$$\begin{aligned} E^G_{[\underline{\mathbf{w}}]} &= \text{probability of misclassification} \\ &= \int P_{(y_T,\underline{\mathbf{x}})}\, e_{(y_T,\underline{\mathbf{x}};\underline{\mathbf{w}})}\, d\underline{\mathbf{x}}\, dy_T \;\stackrel{!}{=}\; \min \qquad \text{(generalization error)} \end{aligned}$$

mathematical expectation

$\Downarrow$ ERM $\Downarrow$ $\leftarrow$ When does this work?

empirical average

$$\begin{aligned} E^T_{[\underline{\mathbf{w}}]} &= \text{fraction of misclassifications} \\ &= \frac{1}{p}\sum_{\alpha=1}^{p} e_{(y_T^{(\alpha)},\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \;\stackrel{!}{=}\; \min \qquad \text{(training error)} \end{aligned}$$

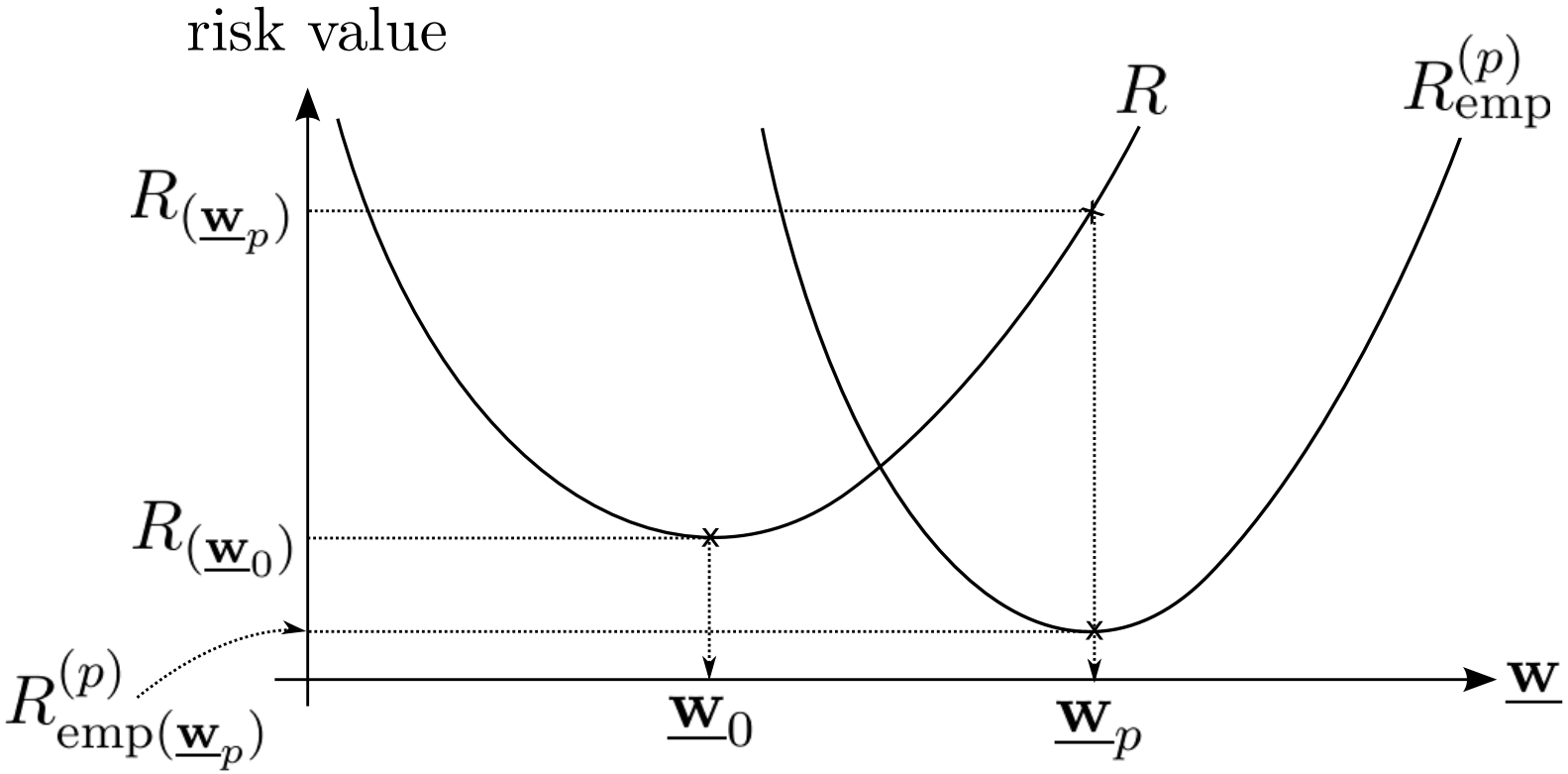
Change in nomenclature ($\rightarrow$ Vapnik, 1998)

probability of misclassification:	$E^G_{[\underline{\mathbf{w}}]}$	$\rightarrow R_{[\underline{\mathbf{w}}]}$	(risk)
fraction of misclassification:	$E^T_{[\underline{\mathbf{w}}]}$	$\rightarrow R_{\mathrm{emp}[\underline{\mathbf{w}}]}$	(empirical risk)

data representation

- p observations $\underline{\mathbf{z}}^{(\alpha)} := (\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)}), \alpha \in \{1, \ldots, p\}$
- observations are drawn **i.i.d.** from an unknown distribution $P(\underline{\mathbf{x}}, y_T) = P(y_T|\underline{\mathbf{x}})\, P(\underline{\mathbf{x}})$.
- **stationarity:** training and test distributions are identical

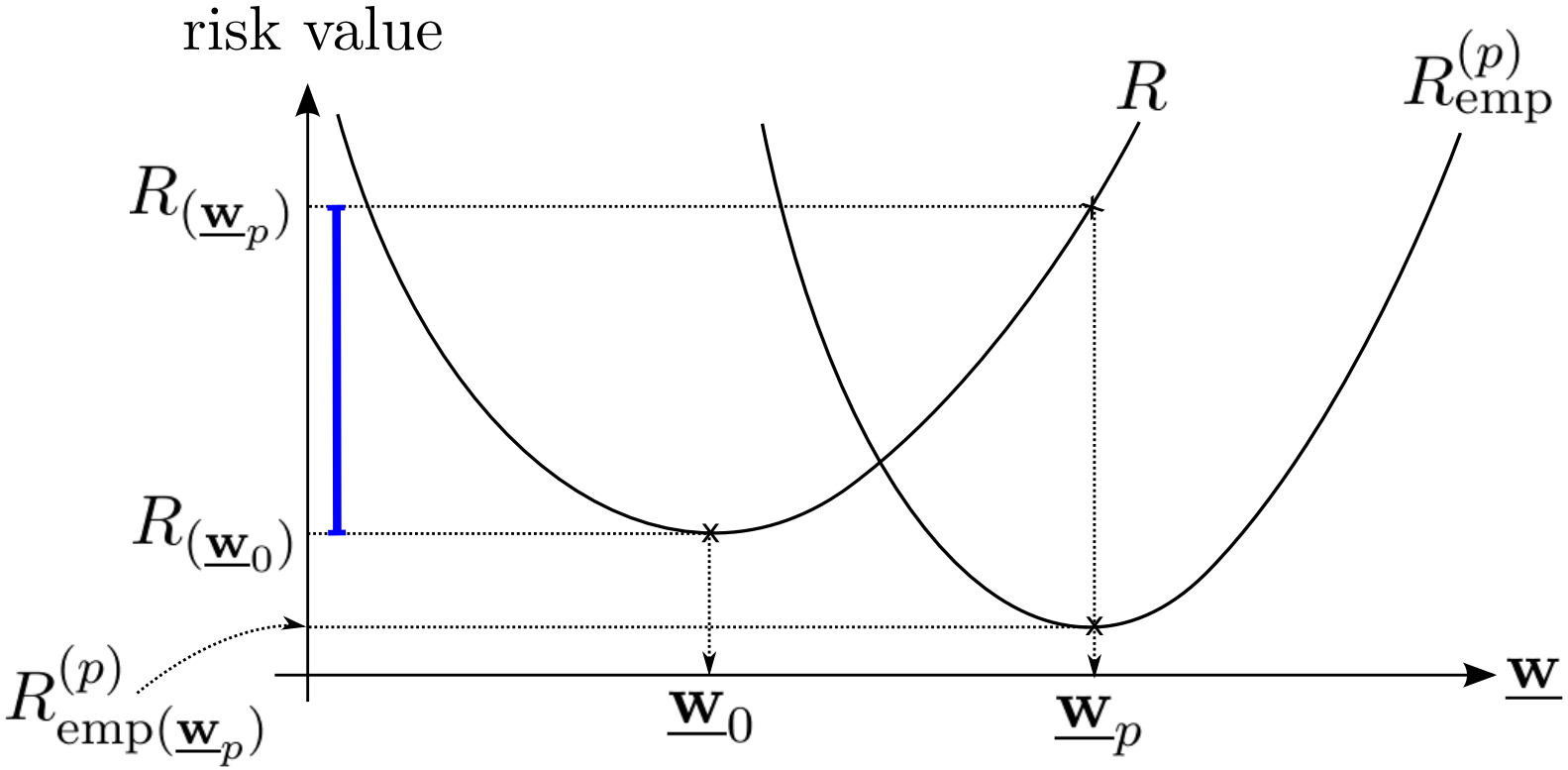
The learning problem



- samples are drawn *i.i.d.* from $P(\underline{\mathbf{x}}, y_T)$

  $\rightsquigarrow$ $\underline{\mathbf{w}}_p$ are random variables

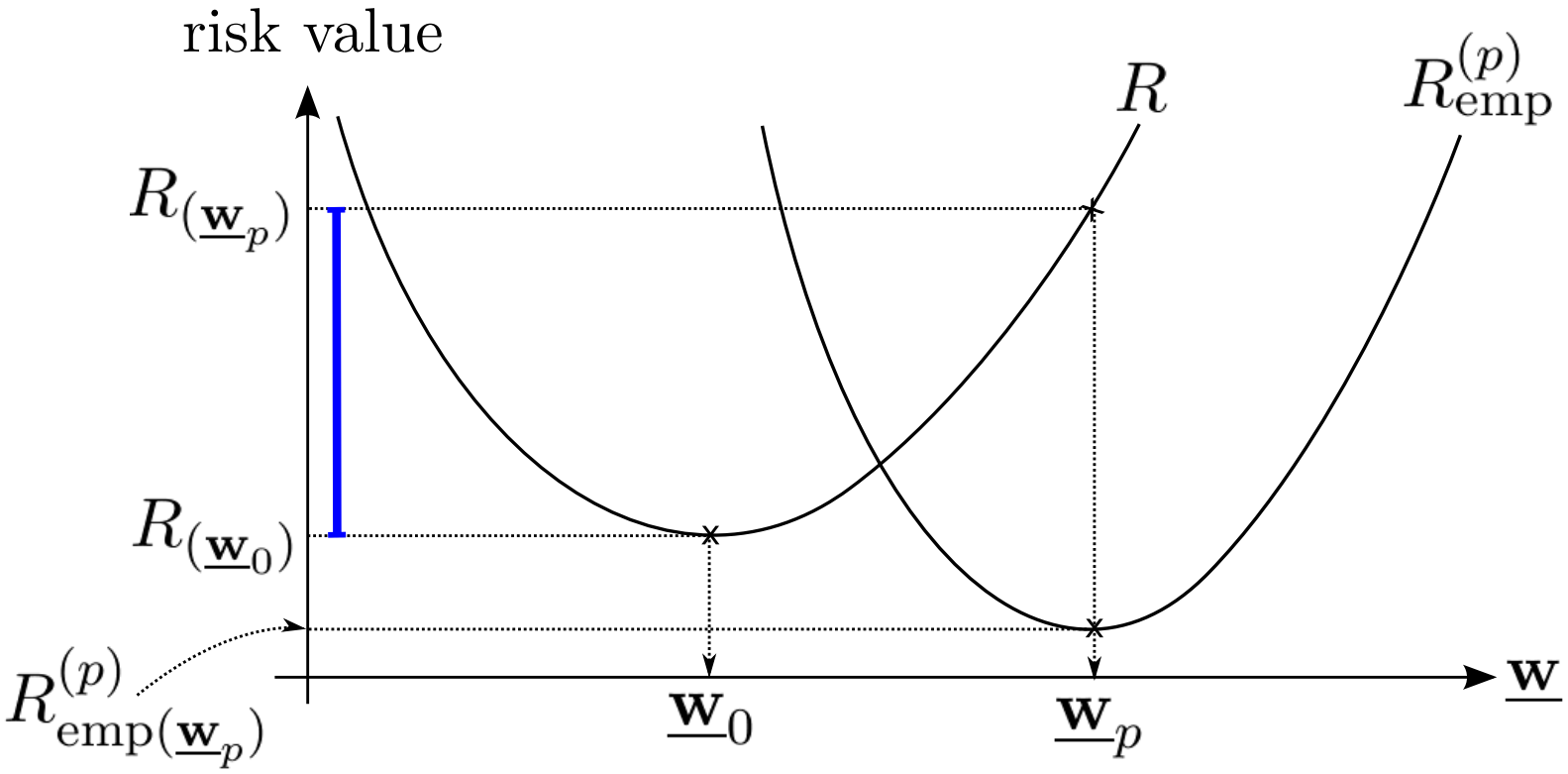
The key questions of Statistical Learning Theory



1. When does inductive learning through ERM work?
   - $R(\underline{\mathbf{w}}_p)$ should reflect the true optimal risk $R(\underline{\mathbf{w}}_0)$ in the limit $p \to \infty$:

$$\lim_{p\to\infty} P\Big\{ \big| R_{(\underline{\mathbf{w}}_p)} - R_{(\underline{\mathbf{w}}_0)} \big| \geq \eta \Big\} \;=\; 0\,, \qquad \text{for all} \quad \eta > 0\,.$$

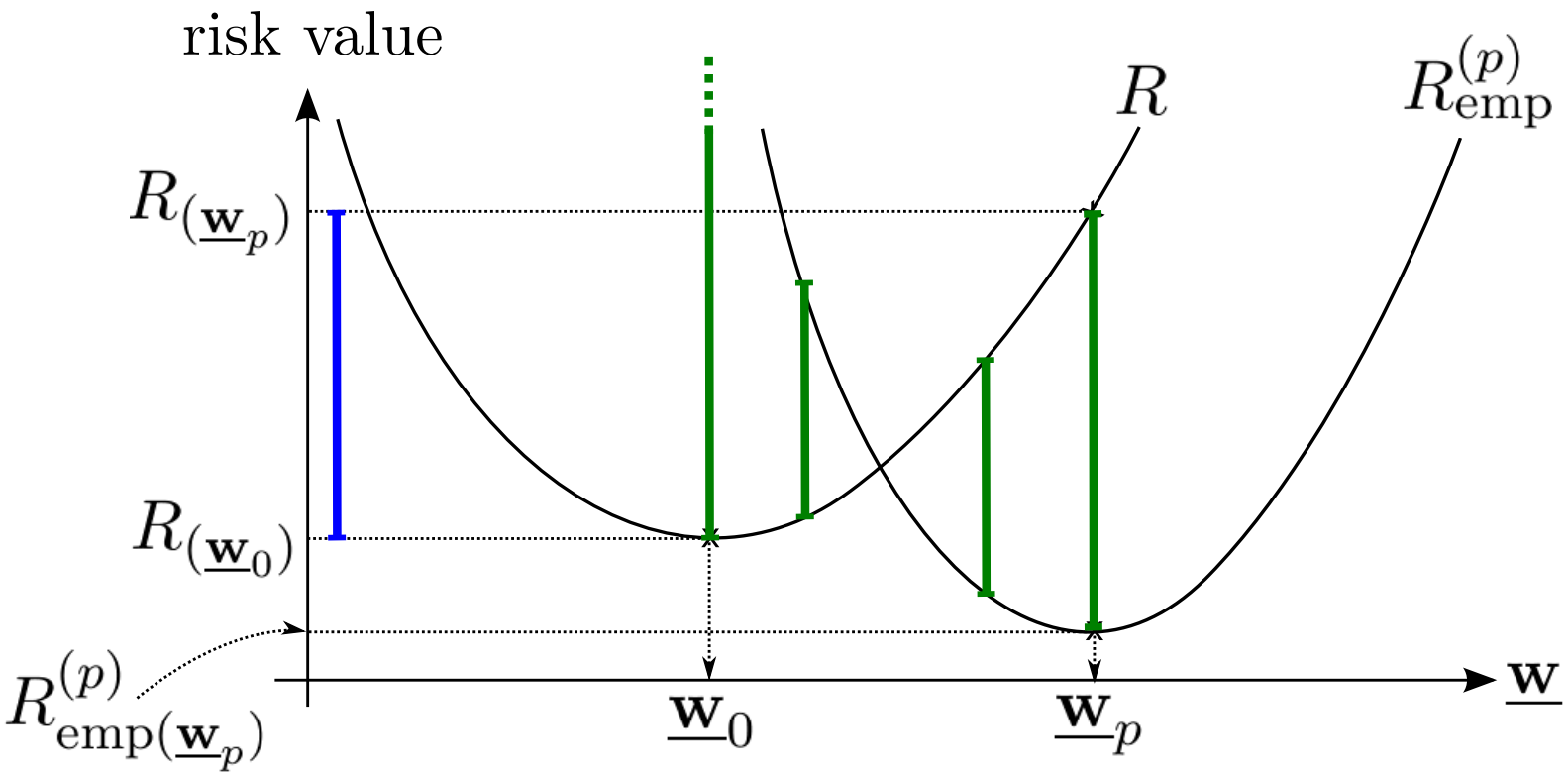
The key questions of Statistical Learning Theory



2. How strongly does $R_{(\underline{\mathbf{w}}_p)}$ differ from $R_{(\underline{\mathbf{w}}_0)}$ for **finite** samples?
   - For a given confidence ϵ, find η such that:

$$P\Big\{\big|R_{(\underline{\mathbf{w}}_p)} - R_{(\underline{\mathbf{w}}_0)}\big| \geq \eta\Big\} \quad < \quad \epsilon\,.$$

The key questions of Statistical Learning Theory



3. Do training errors $R_{\text{emp}(\underline{\mathbf{w}})}$ reflect generalization performance $R_{(\underline{\mathbf{w}})}$?
   - For a confidence ϵ, find η such that:

$$P\Big\{\big|R_{(\underline{\mathbf{w}})} - R_{\text{emp}(\underline{\mathbf{w}})}\big| > \eta\Big\} \;<\; \epsilon\,, \qquad \forall \underline{\mathbf{w}} \in \Lambda\,.$$

2.1.3 General Classifiers: Conditions for Inductive Learning

The growth function $G^{\Lambda}_{(p)}$

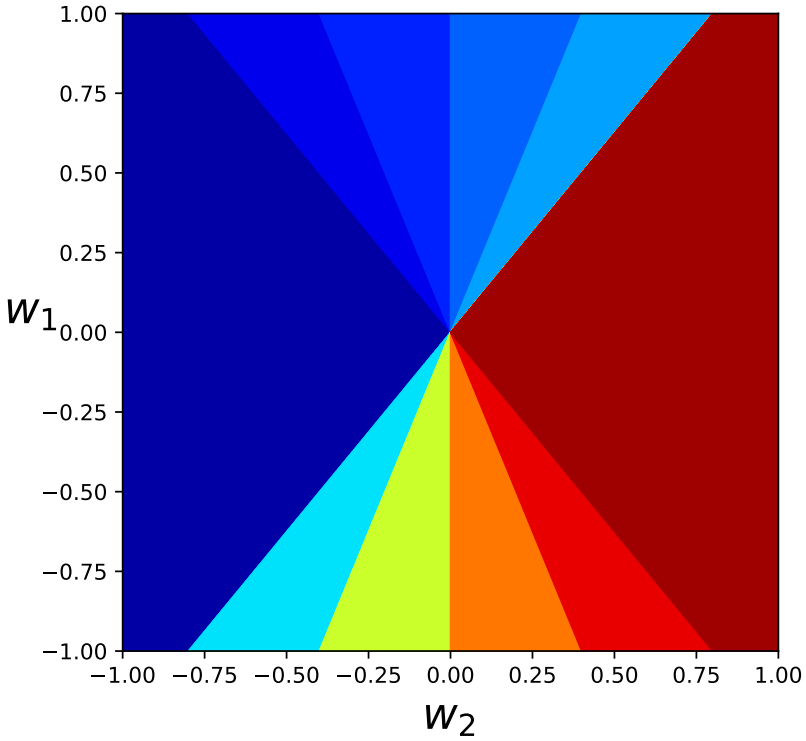
- **data representation:** $\underline{\mathbf{x}} \in \mathbb{R}^N, \quad y_T \in \{-1, +1\}$
- **model class:** set Λ of functions $y_{(\underline{\mathbf{x}};\underline{\mathbf{w}})} \in \{-1, +1\}$
- **binary label vector:** $\underline{\mathbf{y}}_{(\underline{\mathbf{w}})} = \left(y_{(\underline{\mathbf{x}}^{(1)},\underline{\mathbf{w}})}, y_{(\underline{\mathbf{x}}^{(2)},\underline{\mathbf{w}})}, \ldots, y_{(\underline{\mathbf{x}}^{(p)},\underline{\mathbf{w}})}\right)$
  - different classifiers can induce the same label vector on the training set
- **number** of different vectors $\underline{\mathbf{y}}_{(\underline{\mathbf{w}})}$ induced by all $\underline{\mathbf{w}} \in \Lambda$:

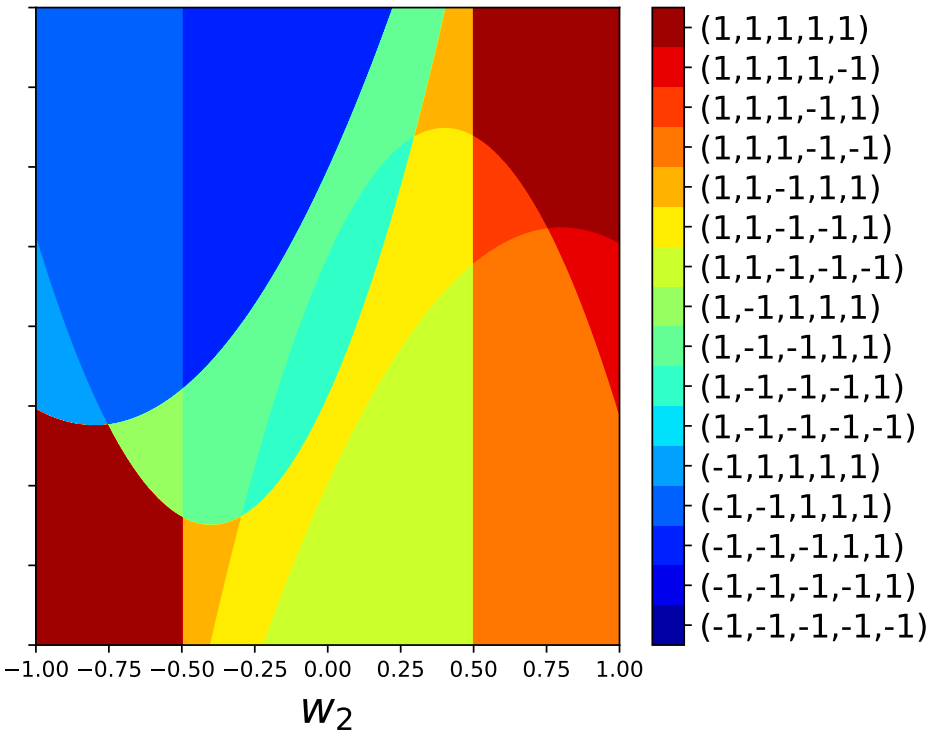
$$N^{\Lambda}_{(\underline{\mathbf{x}}^{(1)},\ldots,\underline{\mathbf{x}}^{(p)})} \leq 2^p \qquad \text{(depends on } \Lambda \text{ and the sample)}$$

The growth function $G^{\Lambda}_{(p)}$

$$5 \quad \text{data points} \quad x^{(i)} \in \{-0.8, -0.4, 0, 0.4, 0.8\}$$

$$f(x|\underline{\mathbf{w}}) = \text{sign}\left(w_1\, x + w_2\right) \qquad f(x|\underline{\mathbf{w}}) = \text{sign}\left(w_1\, x + (x - w_2)^2 - \tfrac{1}{4}\right)$$





$$N^{\Lambda}_{(x^{(1)},\ldots,x^{(5)})} = 10 \leq 2^5 \qquad N^{\Lambda}_{(x^{(1)},\ldots,x^{(5)})} = 13 \leq 2^5$$

The growth function $G^{\Lambda}_{(p)}$

$$G^{\Lambda}_{(p)} = \ln \underbrace{\left(\sup_{\underline{\mathbf{x}}^{(1)},\ldots\underline{\mathbf{x}}^{(p)}} N^{\Lambda}_{(\underline{\mathbf{x}}^{(1)},\ldots,\underline{\mathbf{x}}^{(p)})} \right)}_{\text{measure independent of the specific sample}}$$

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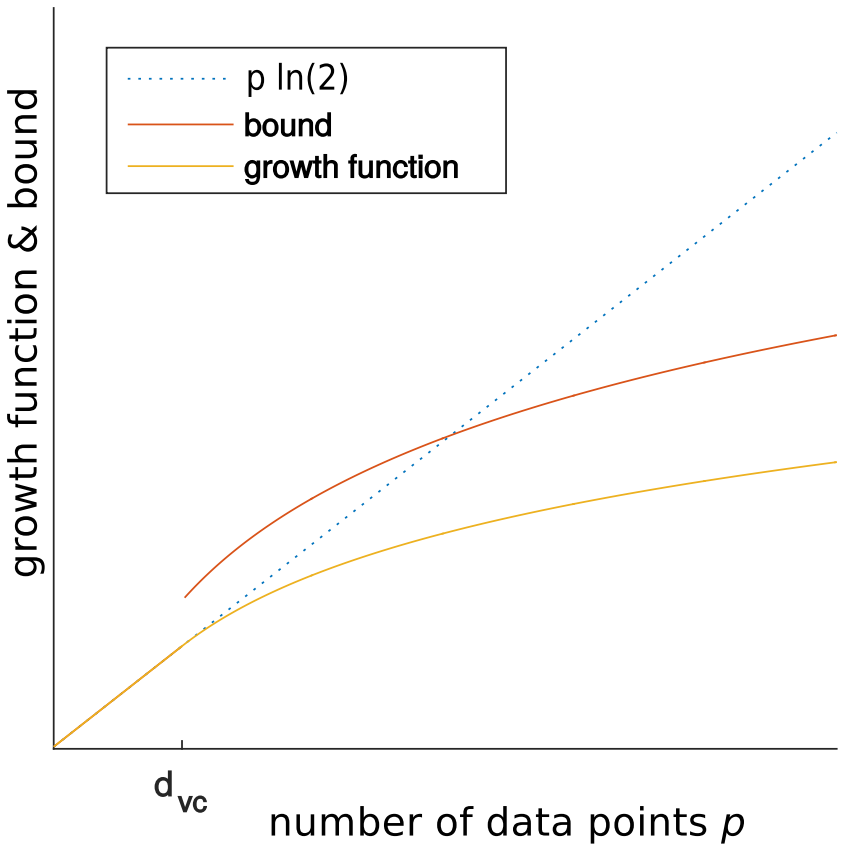
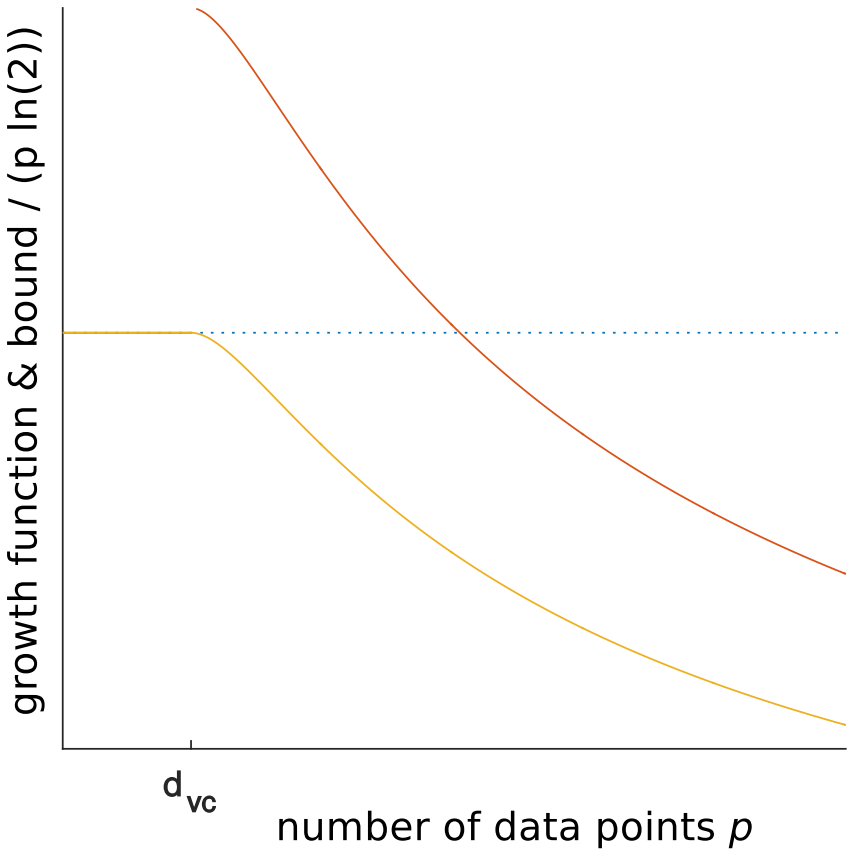
- bound on growth function (Vapnik, 1998)

$$G^{\Lambda}_{(p)} \begin{cases} = p \ln 2 & \text{for } p \leq \mathrm{d_{VC}} \\ \leq \mathrm{d_{VC}} \left(1 + \ln \frac{p}{\mathrm{d_{VC}}}\right) & \text{for } p > \mathrm{d_{VC}} \end{cases}$$

- Vapnik-Chernovenkis dimension d_{VC}:
  capacity measure of the model class

The growth function $G^{\Lambda}_{(p)}$





d_{VC}: capacity measure of the model class

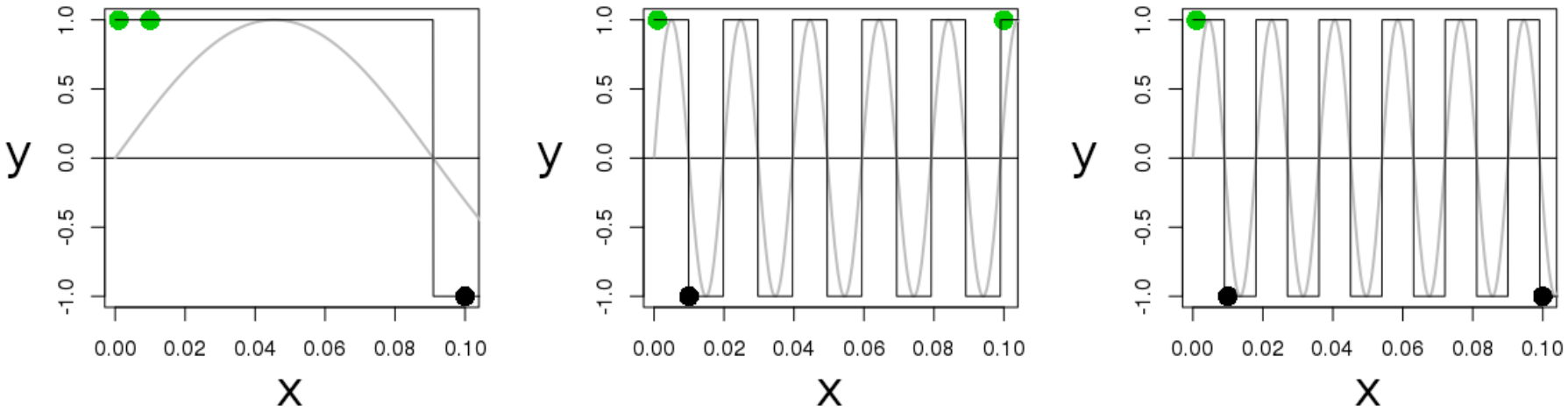
The Vapnik-Chervonenkis dimension

- $\mathrm{d_{VC}}$ is ...
  - largest number of data points on which (at least for one set) 2^p different loss vectors (i.e. label assignments) can be induced
  - capacity measure for a given set of models
  - purely combinatorial concept (worst case scenario)
  - provides a bound on the growth function ...
  - ... $\rightsquigarrow$ determines the number of samples needed to reliably learn as we will see

- $\mathrm{d_{VC}}$ is typically hard to compute
  - bounds have been derived for some architectures
  - e.g. MLPs with sigmoidal units: $\mathrm{d_{VC}} = \mathcal{O}(n^2)$, where n is number of parameters, Koiran and Sontag (1995)

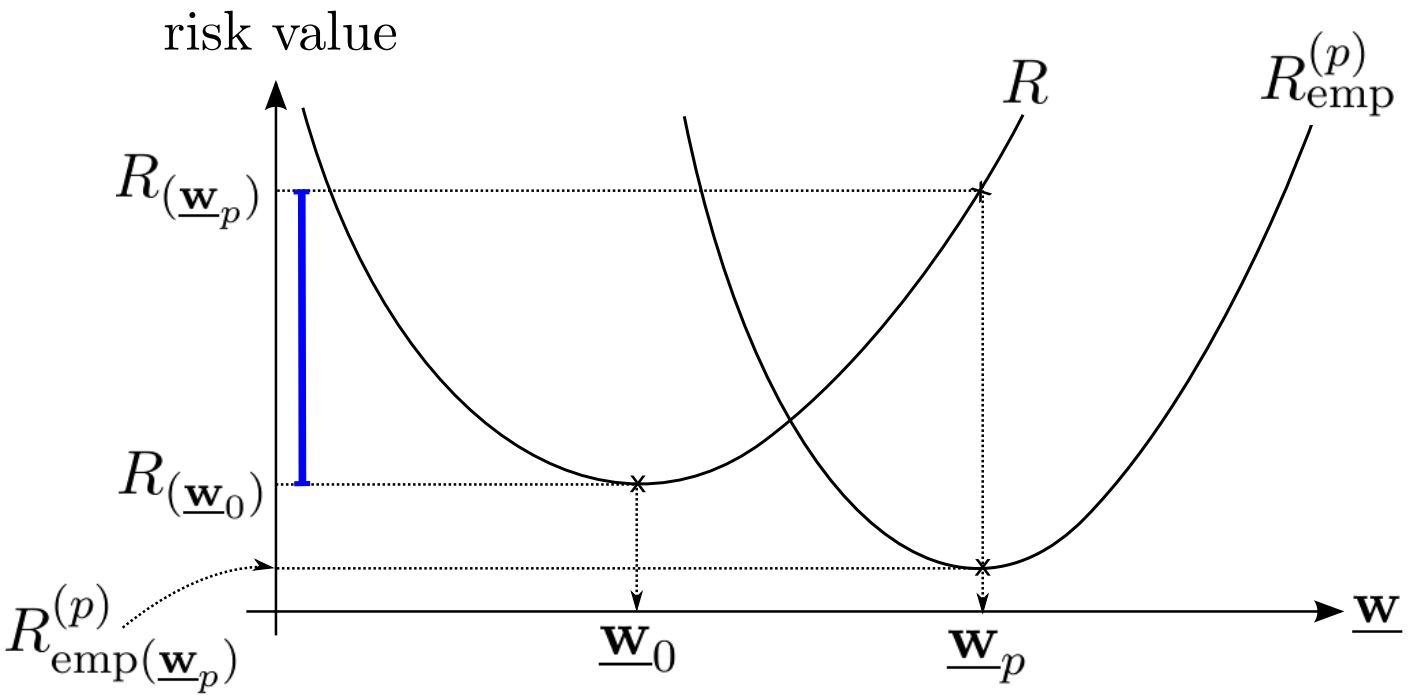
An unlearnable problem

- set of classifiers: $y(x;a) = \mathrm{sign}[\sin(ax)]$, single model parameter a
- selection of classifiers (cf. Vapnik, 1998)
  - take $x^{(i)} = 10^{-i},\ i = 1, ..., p$, and assign any $y^{(i)} \in \{-1, +1\}$
  - set $a = \pi\left(1 + \sum_{i=1}^{p} \frac{(1-y^{(i)})10^i}{2}\right)$



- a single model parameter, but an infinite d_{VC}

The solution to learning problem 1

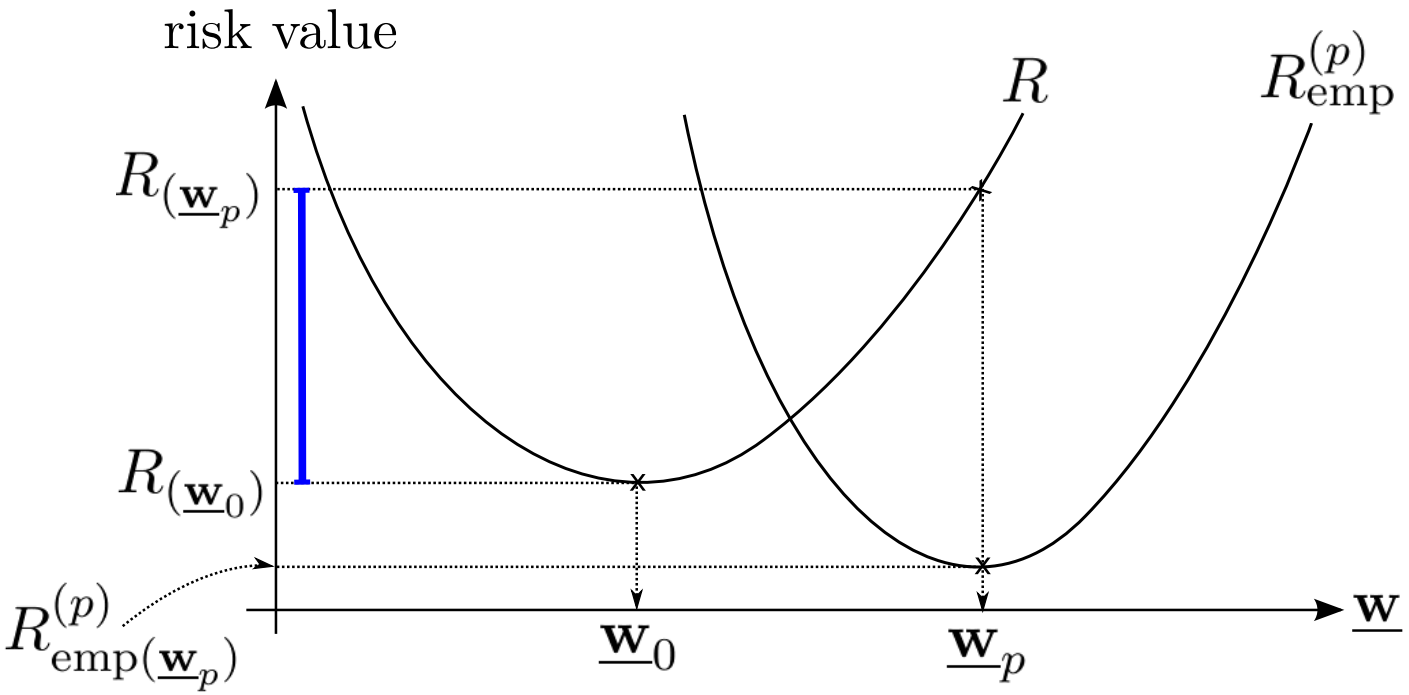


1. **learnability**: asymptotic result

$$\mathrm{d_{VC}} \text{ finite} \quad \Rightarrow \quad \lim_{p\to\infty} P\left\{ \left| R_{(\underline{\mathbf{w}}_p)} - R_{(\underline{\mathbf{w}}_0)} \right| \geq \eta \right\} \;=\; 0\,, \quad \forall \eta > 0$$

- convergence of ERM is guaranteed if $\mathrm{d_{VC}}$ is finite

The solution to learning problem 2

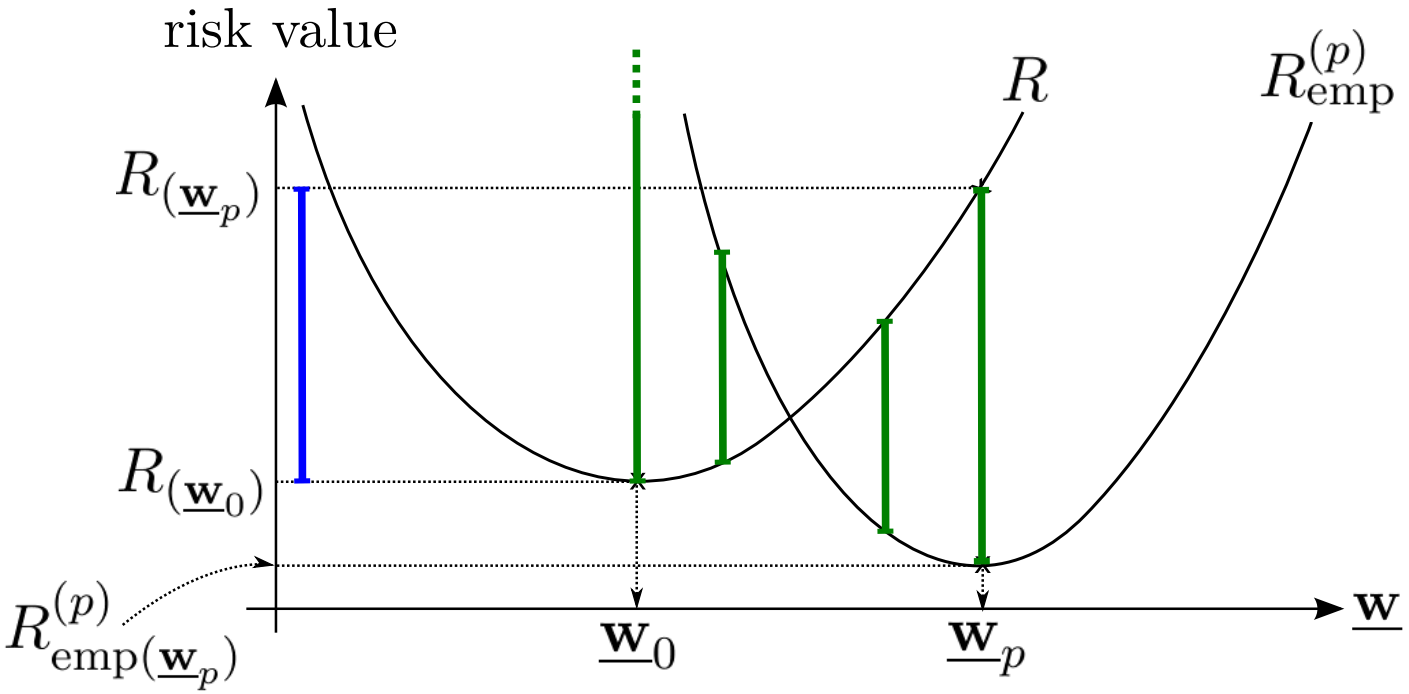


2. **finite samples:** deviation from the optimal model

$$P\left\{\left|R_{(\underline{\mathbf{w}}_p)} - R_{(\underline{\mathbf{w}}_0)}\right| > \left(\frac{G^{\Lambda}_{(2p)} - \ln\frac{\epsilon}{8}}{p}\right)^{\frac{1}{2}} + \left(-\frac{\ln\frac{\epsilon}{2}}{2p}\right)^{\frac{1}{2}} + \frac{1}{p}\right\} < \epsilon$$

- finite d_{VC}: increased sample size $p \Rightarrow$ reduction of the bound

The solution to learning problem 3



3. **finite samples:** bound on the generalization error

$$P\left\{ \sup_{\underline{\mathbf{w}}\in\Lambda} \left| R_{(\underline{\mathbf{w}})} - R^{(p)}_{\mathsf{emp}(\underline{\mathbf{w}})} \right| > \eta \right\} < 4\exp\left(G^{\Lambda}_{(2p)} - p\left(\eta - \tfrac{1}{p}\right)^2 \right)$$

- bound non-trivial only if $G^{\Lambda}_{(2p)}$ is sub-linear in p (see blackboard)

References I

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End of Section 2.1

the following slides contain

OPTIONAL MATERIAL

Mathematical preliminaries

binomial coefficient

$$\binom{n}{k} = \begin{cases} \frac{n!}{k!(n-k)!}, & n \geq k \\ 0, & \text{else} \end{cases}$$

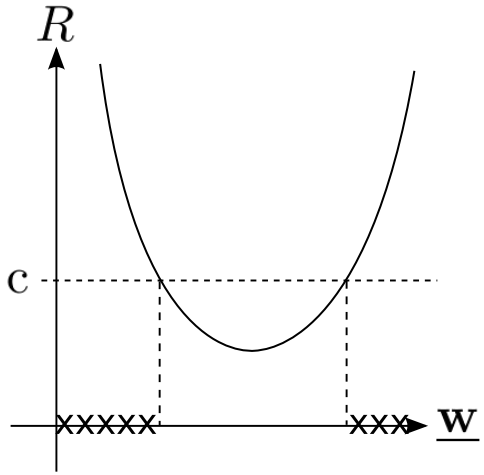
binomial distribution

$$f(k|n,p) = \binom{n}{k} q^k (1-q)^{n-k}$$

cumulative distribution

$$F(k|n,p) = Pr(X \leq k) = \sum_{i=0}^{k} f(i|n,p)$$

Strict consistency

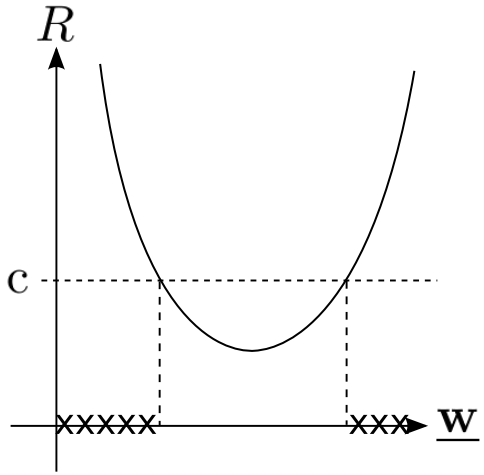


$\Lambda = \{\underline{\mathbf{w}}\}$: set of all predictors

$\Lambda_{(c)} \subset \Lambda$: set of all "bad" predictors

$\times\times\times$: $\Lambda_{(c)} = \{\underline{\mathbf{w}} : R_{(\underline{\mathbf{w}})} \geq c\}$

Strict consistency



$\Lambda = \{\underline{\mathbf{w}}\}$: set of all predictors

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$\times\times\times$: $\Lambda_{(c)} = \{\underline{\mathbf{w}} : R_{(\underline{\mathbf{w}})} \geq c\}$

definition and implications of **strict consistency**

$$\lim_{p\to\infty} P\left\{ \left| \inf_{\underline{\mathbf{w}}\in\Lambda_{(c)}} R^{(p)}_{\mathrm{emp}(\underline{\mathbf{w}})} - \inf_{\underline{\mathbf{w}}\in\Lambda_{(c)}} R_{(\underline{\mathbf{w}})} \right| \geq \eta \right\} = 0$$

strict consistency $\overset{\rightarrow}{\not\leftarrow}$ inductive learning via ERM

proof: supplementary material

Strict Consistency

key theorem of learning theory

Let $a \leq R_{(\underline{\mathbf{w}})} \leq A$ for all $\underline{\mathbf{w}} \in \Lambda$ and $P_{(\underline{\mathbf{z}})} \in \Pi$, then:

the ERM procedure is strictly consistent for Λ and Π

$$\Updownarrow$$

$$\lim_{p \to \infty} P\left\{ \sup_{\underline{\mathbf{w}} \in \Lambda} \left(R_{(\underline{\mathbf{w}})} - R^{(p)}_{\mathrm{emp}(\underline{\mathbf{w}})} \right) > \eta \right\} = 0\,, \quad \text{for all} \quad P_{(\underline{\mathbf{z}})} \in \Pi\,.$$

proof: supplementary material

- uniform (one-sided) convergence of $R^{(p)}_{\mathrm{emp}(\underline{\mathbf{w}})}$ to $R_{(\underline{\mathbf{w}})}$ for $p \to \infty$